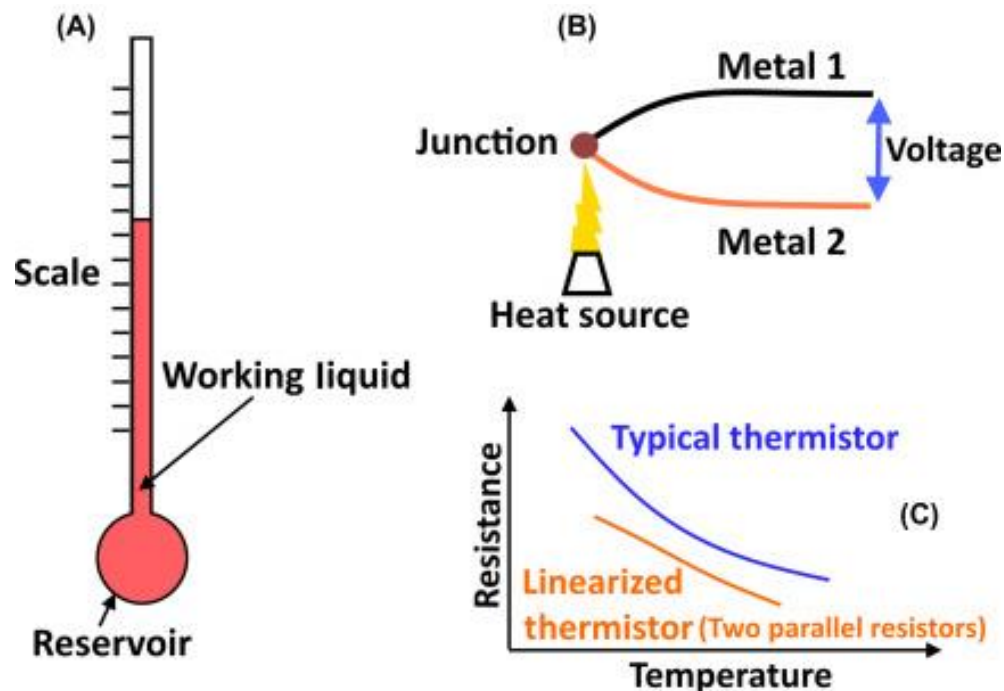


Introduction to Measurements



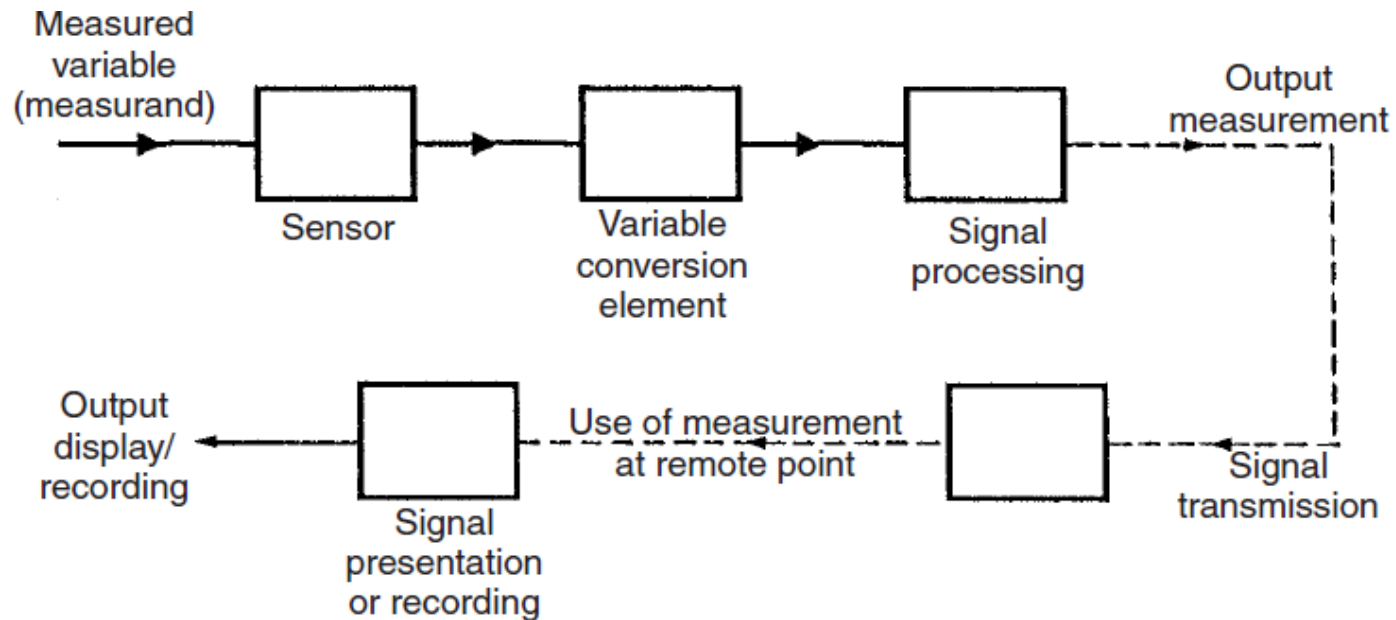
Introduction to Measurement

- A sensor is a device that transfers the measured physical property into some output signal that can be processed or displayed.
- Sensors use mostly electrical output signals.

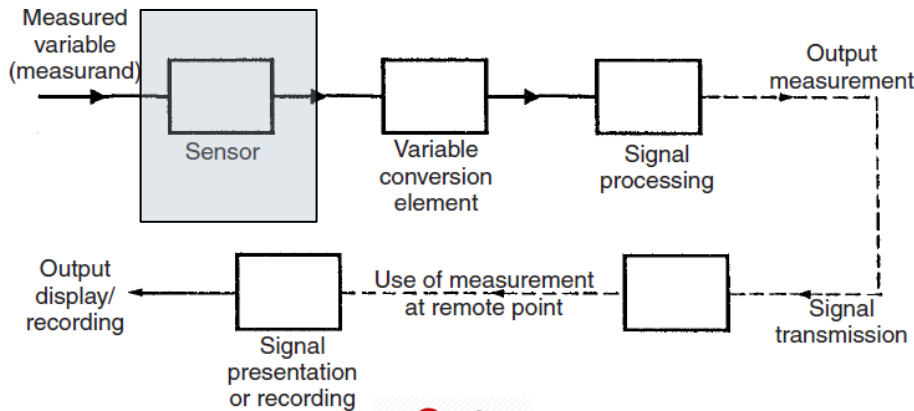


Elements of a measurement system

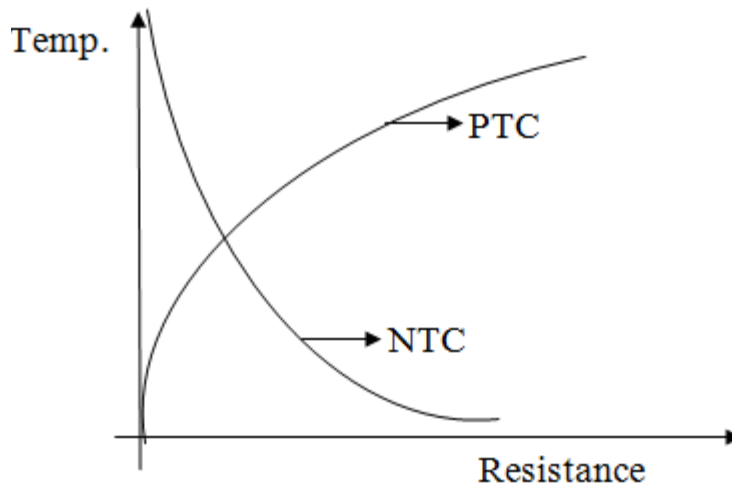
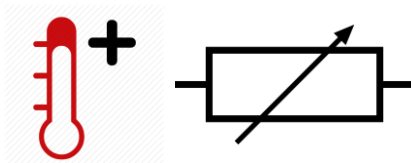
- In simple cases, the system can consist of only a single unit that gives an output reading or signal. However, the sensor is usually placed in a measuring chain:



Elements of a measurement system



Sensors gives an output that is a function of the measurand (the input applied to it).



Tensile force

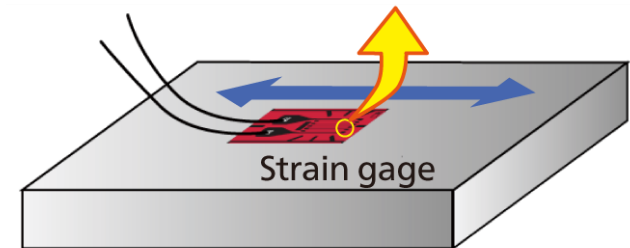


reduce the cross-sectional area $\rightarrow$ increase the resistance

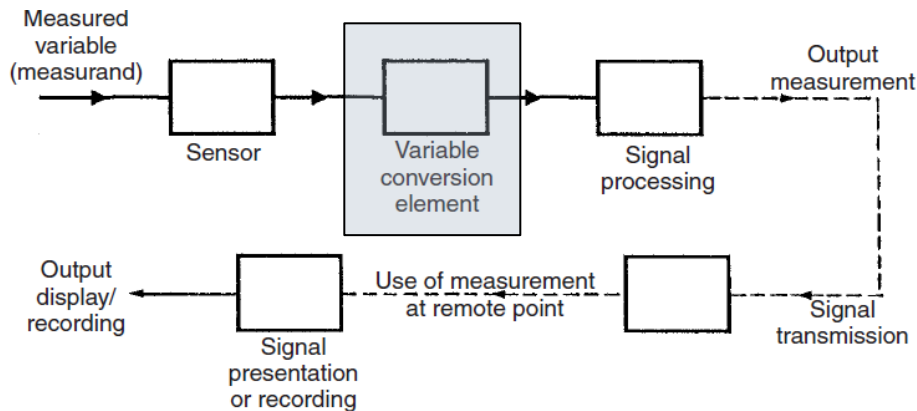
Compression force



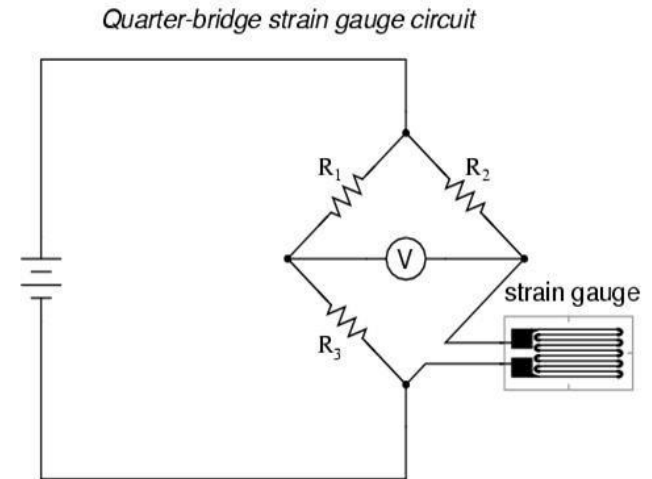
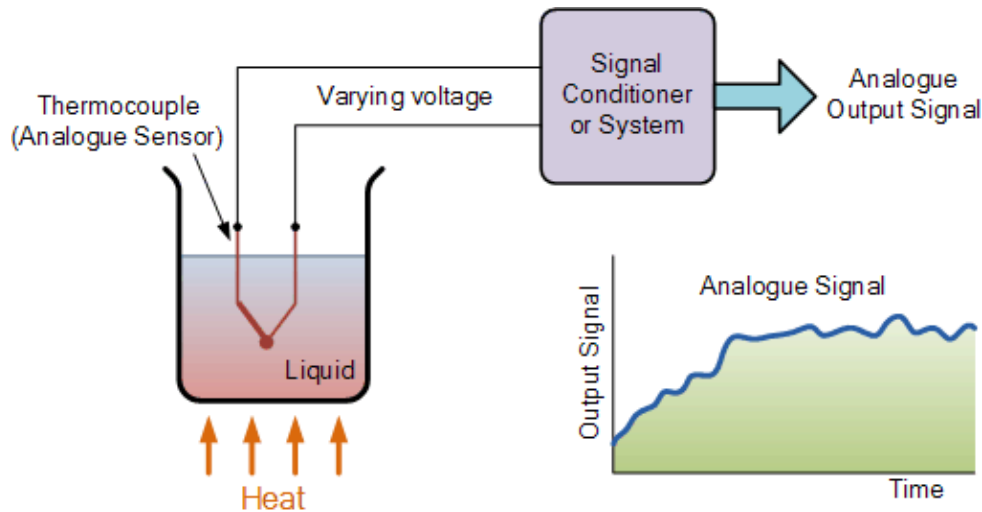
increase the cross-sectional area $\rightarrow$ reduce the resistance



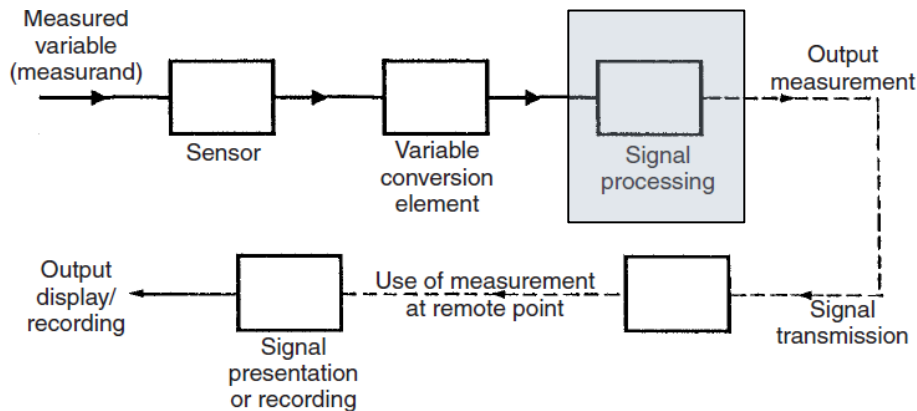
Elements of a measurement system



- **Variable conversion elements** convert the output variable into a more convenient form.
- In some cases, the primary sensor and variable conversion element are combined, and known as a ***transducer***.



Elements of a measurement system

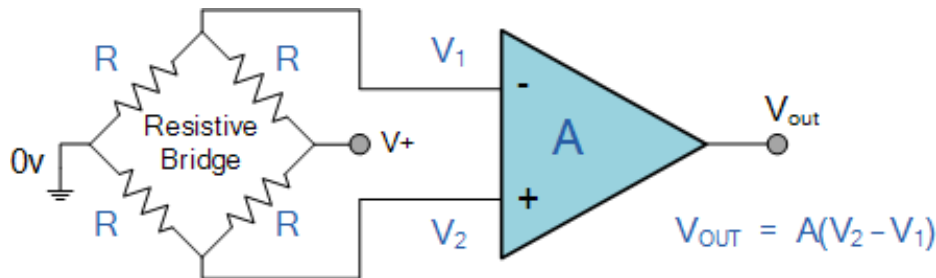


□ Signal processing elements improve the quality of the output of a measurement system.

■ **Electronic amplifier**

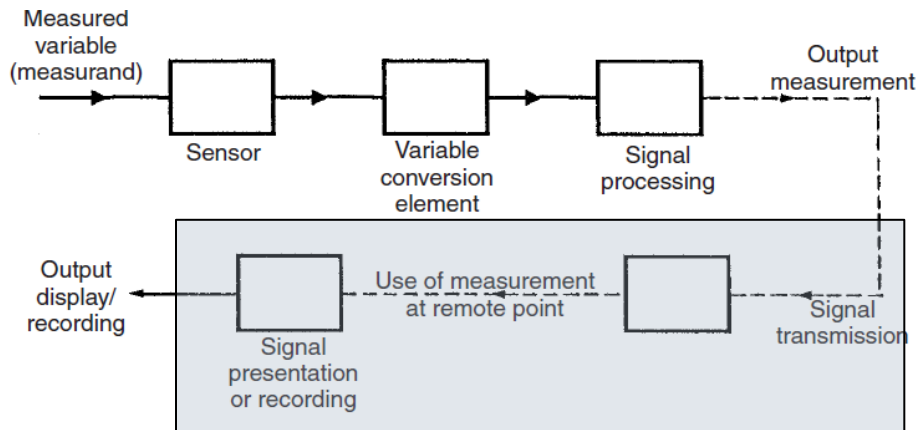
■ **Filter**

□ In some devices, signal processing is incorporated into a transducer, which is then known as a **transmitter**.

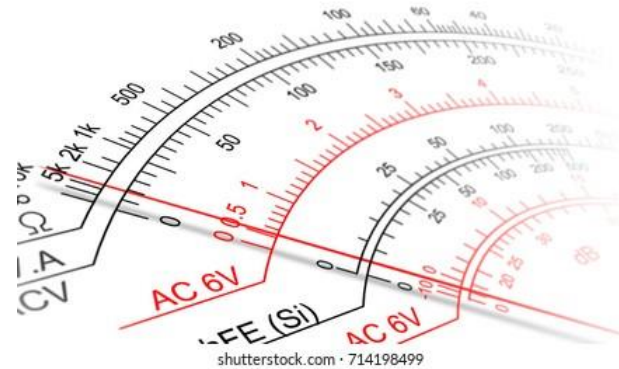


In some cases, the word 'sensor' is used generically to refer to both transducers and transmitters

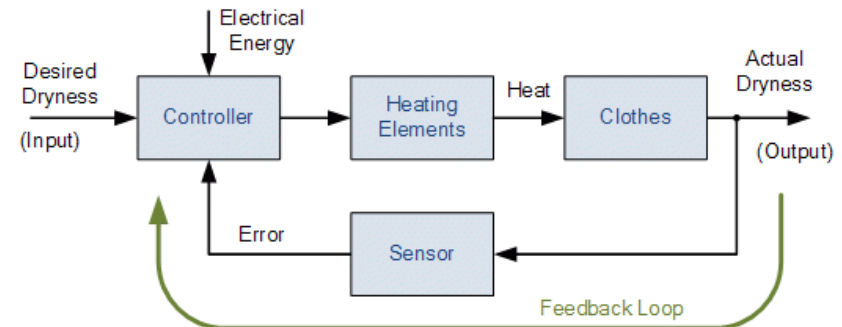
Elements of a measurement system



□ Signal presentation or recording

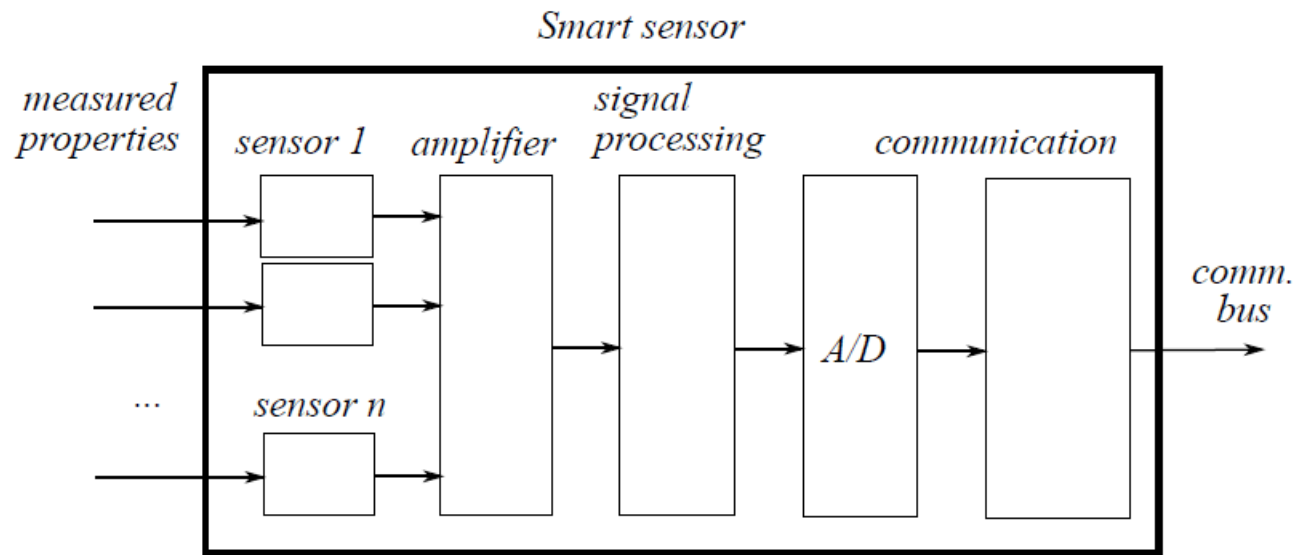


□ The signal transmission element



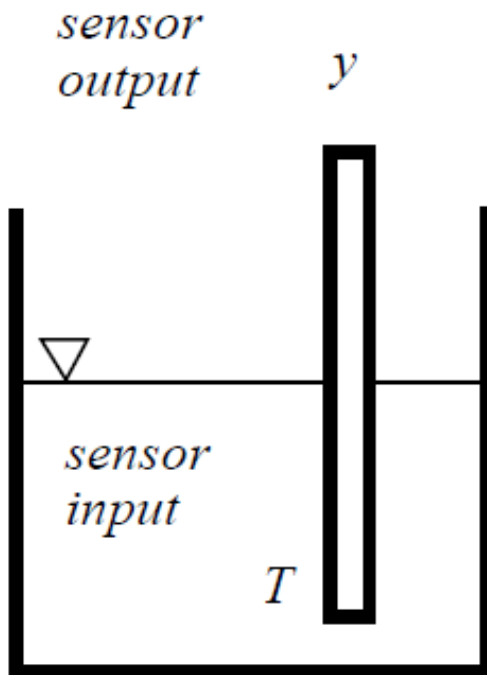
Elements of a measurement system

- ❑ In smart sensors, all or some parts of the measuring chain are integrated in an independent block (can be integrated circuit, circuit board).
- ❑ Smart sensors are most commonly composed of the main sensor (or sensors), amplifiers, analog/digital converters, microprocessor and some communication interface.



Static Properties

- ❑ **Experiment:** The water temperature in a tank is measured with a temperature sensor.



- ❑ Sensor properties can be described in transients (functions of time) or in steady state (not functions of time).
- ❑ Steady state is the state in which **BOTH input signal AND output signal are constant.**

Accuracy and inaccuracy

- The accuracy of an instrument is a measure of how close the output reading of the instrument is to the correct value.
- **Absolute error** Δy is the difference between the measured value y_m and correct value y_c of the measured property.

$$\Delta y = y_m - y_c$$

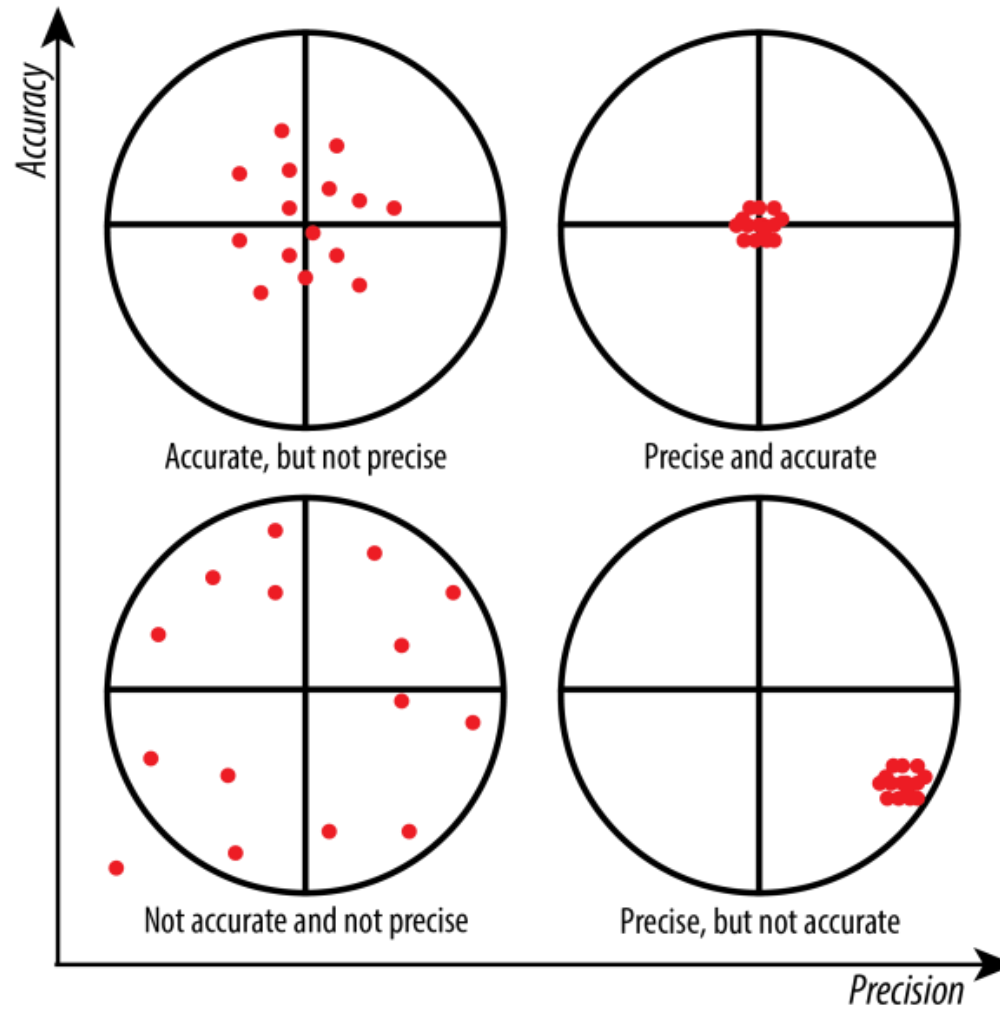
- **Relative error** $\delta(y)$ is the ratio between absolute error Δy to the instantaneously measured value y_m .

$$\delta(y) = \frac{\Delta y}{y_m} \times 100\%$$

Precision/repeatability/reproducibility

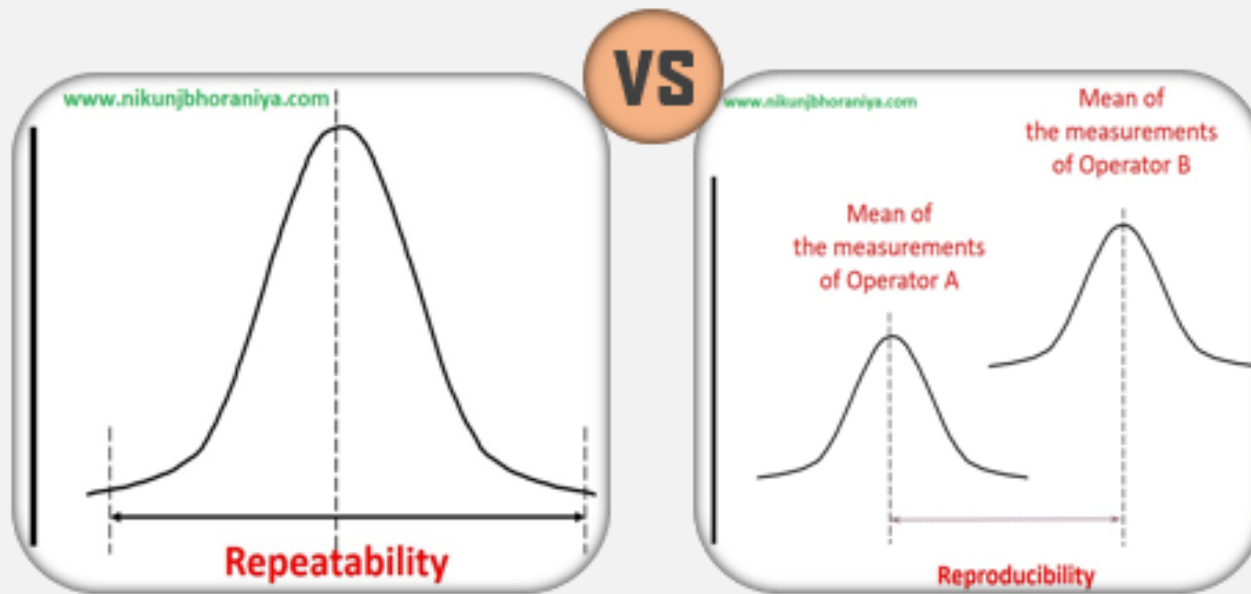
- ❑ **Precision** is a term that describes an instrument's degree of freedom from random errors.
- ❑ **Precision $\neq$ Accuracy**
- ❑ Low accuracy measurements from a high precision instrument are normally caused by a bias in the measurements.

Precision/repeatability/reproducibility



Precision/repeatability/reproducibility

REPEATABILITY vs REPRODUCIBILITY



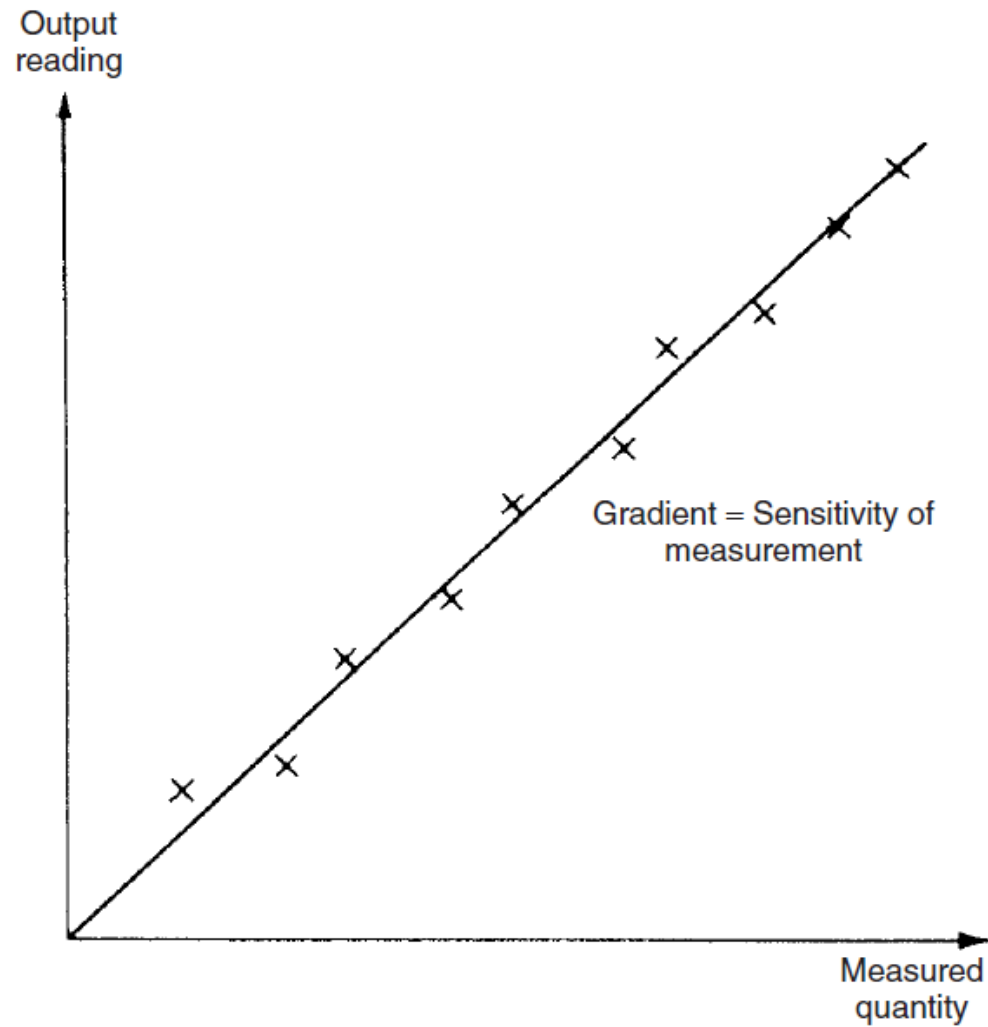
Precision/repeatability/reproducibility

- **Repeatability:** the closeness of output readings when the same input is applied repetitively over a short period of time, with the **same** measurement conditions, same instrument and observer, same location and same conditions of use maintained throughout.
- **Reproducibility:** the closeness of output readings for the same input when **there are changes** in the method of measurement, observer, measuring instrument, location, conditions of use and time of measurement.

Precision/repeatability/reproducibility

- **Tolerance** describes the maximum deviation of a manufactured component from some specified value.
 - One resistor have a nominal value 1000Ω and tolerance 5% might have an actual value anywhere between 950Ω and 1050Ω .

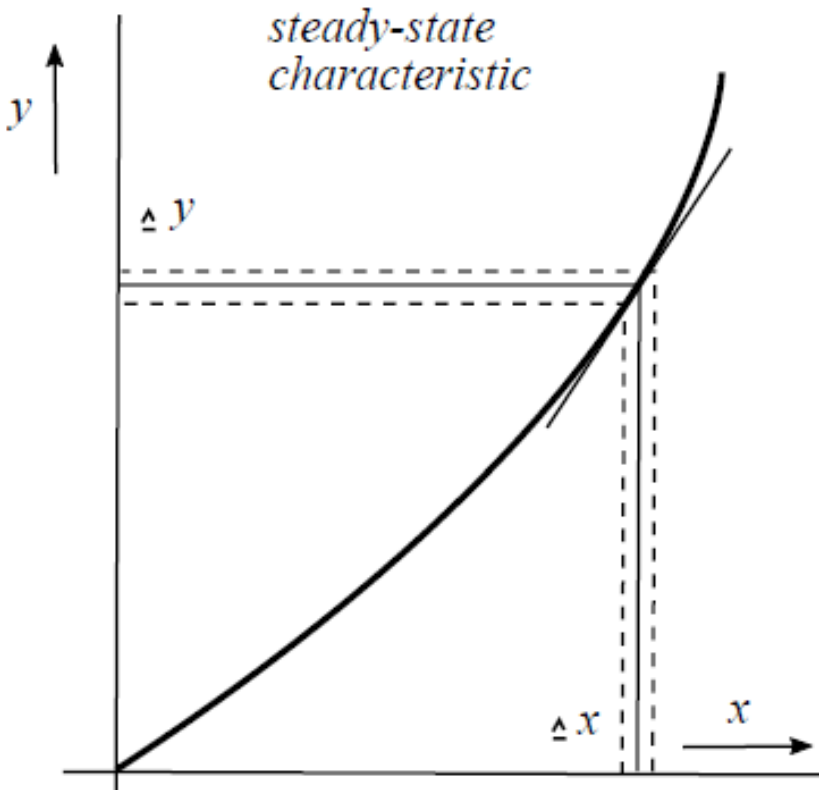
Linearity



Sensitivity of measurement

- The **sensitivity** of measurement is a measure of the change in instrument output that occurs when the quantity being measured changes by a given amount.

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$



- Example: a pressure of 2 bar produces a deflection of 10 degrees in a pressure transducer, the sensitivity of the instrument is 5 degrees/bar

Sensitivity of measurement

- **Example:** The following resistance values of a platinum resistance thermometer were measured at a range of temperatures. Determine the measurement sensitivity of the instrument in ohms/°C.

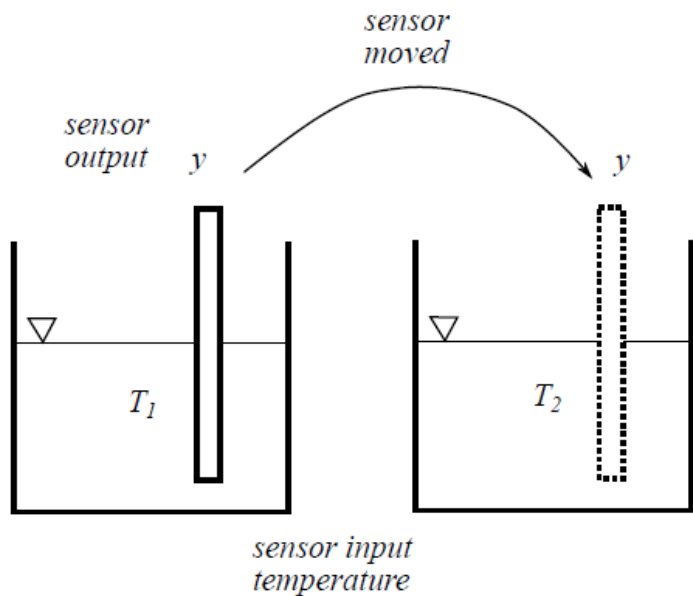
Resistance (Ω)	Temperature ($^{\circ}\text{C}$)
307	200
314	230
321	260
328	290

Threshold and Resolution

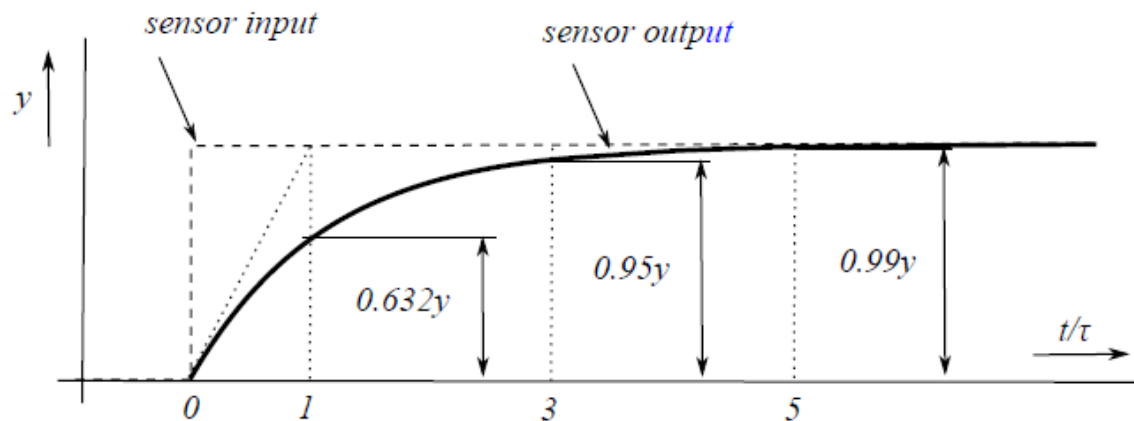
- ❑ **Threshold:** the minimum level of input that causes the change in the instrument output reading to be of a large enough magnitude to detectable.
- ❑ **Resolution:** the lower limit on the magnitude of the change in the input measured quantity that produces an observable change in the instrument output.

Dynamic Properties

□ **Transient responses:** the transient response is the response of a system to a step change on its input.



□ The sensor input signal change with a step. The output signal, however, will not be a step but a transient function.



Dynamic Properties

- Based on the system properties it will take a shorter or longer time before a new steady state is reached.
- In any linear, time-invariant measuring system, the following general relation can be written between input and output for time $(t) > 0$:

$$\frac{a_n(d^n q_0)}{dt^n} + a_{n-1} \frac{d^{n-1} q_0}{dt^{n-1}} + \dots + a_1 \frac{dq_0}{dt} + a_0 q_0 = b_0 q_i$$

Where q_i is the measured quantity, q_0 is the output reading and $a_0 \dots a_n, b_0$ are constants.

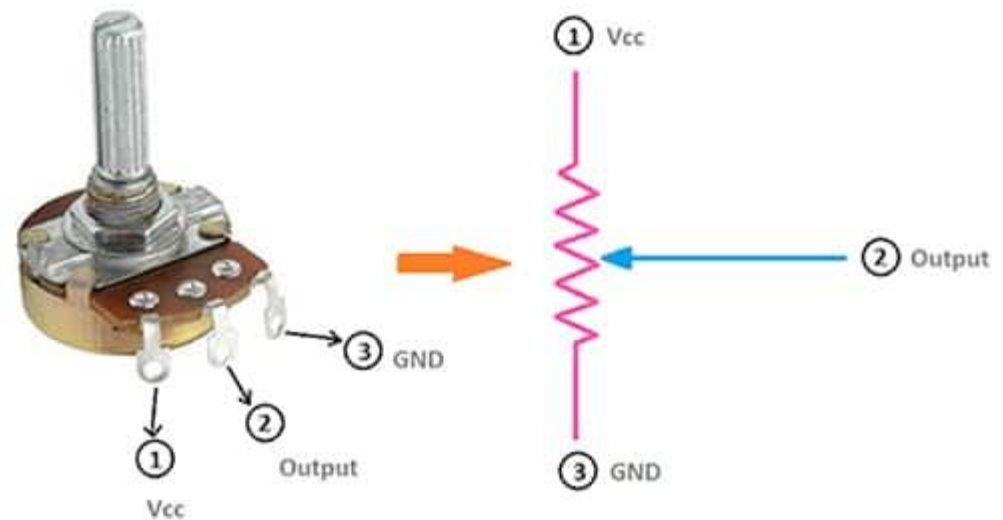
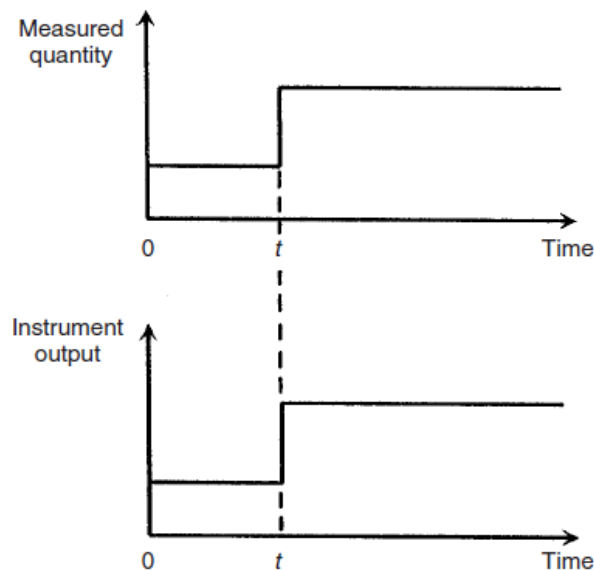
Dynamic Properties

□ Zero order instrument

If all the coefficients $a_1 \dots a_n$ except for a_0 are assumed zero, then:

$$a_0 q_0 = b_0 q_i \text{ or } q_0 = \frac{b_0 q_i}{a_0} = K q_i$$

Where K is the instrument sensitivity.



Dynamic Properties

□ First order instrument

If all the coefficients $a_2 \dots a_n$ except a_0, a_1 for are assumed zero, then:

$$\frac{a_1 dq_0}{dt} + a_0 q_0 = b_0 q_i$$
$$\frac{a_1}{a_0} \times \frac{dq_0}{dt} + q_0 = \frac{b_0}{a_0} \times q_i$$

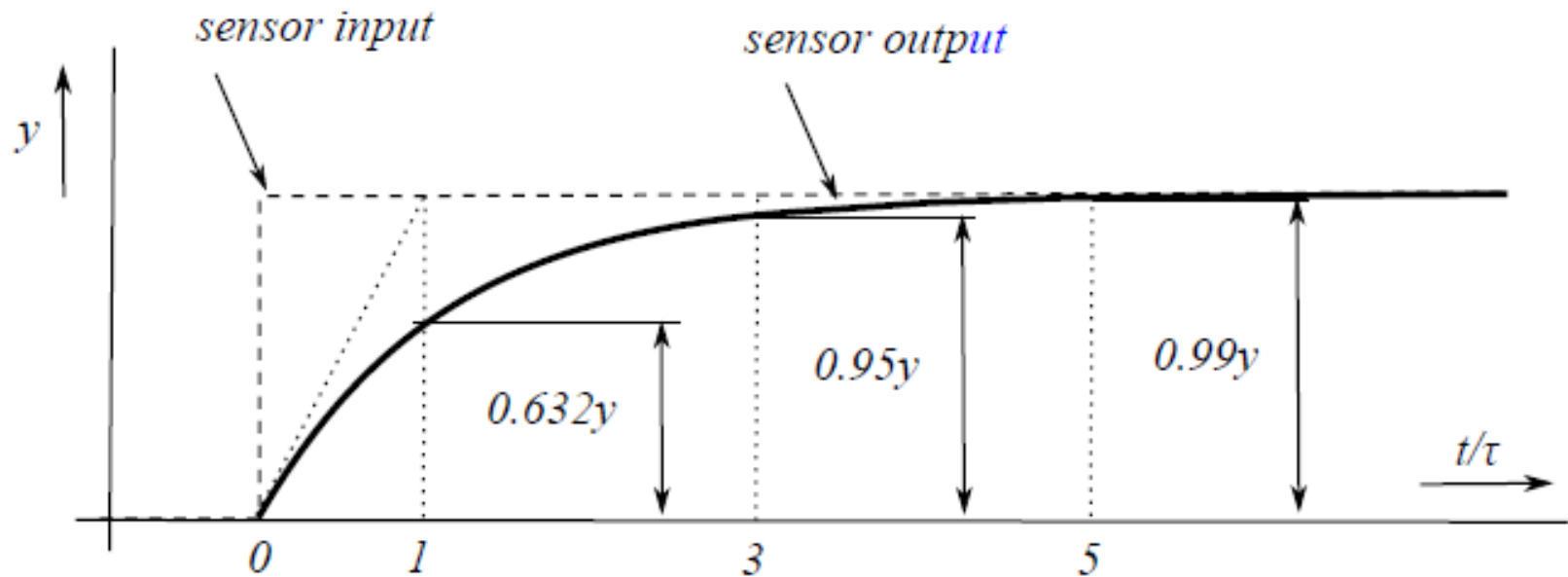
Defining:

- $K = \frac{b_0}{a_0}$ as the static sensitivity/gain – system behavior in steady state
- $\tau = a_1/a_0$ as the time constant of the system – how fast the system can respond to changes

Dynamic Properties

□ First order instrument

- The time constant τ of the step response is the time taken for the output quantity q_0 to reach 63% of its final value.
- 5τ is considered the steady state with sufficient accuracy.



Dynamic Properties

Ví dụ:

Khí cầu được trang bị các dụng cụ đo nhiệt độ và độ cao và có thiết bị vô tuyến có thể truyền các kết quả đầu ra của các thiết bị này trở lại mặt đất. Khí cầu ban đầu được neo vào mặt đất với các số đọc đầu ra của thiết bị ở trạng thái ổn định. Dụng cụ đo độ cao là thiết bị bậc 0 và bộ chuyển đổi nhiệt độ bậc nhất với hằng số thời gian là 15 giây. Nhiệt độ mặt đất là $T_0 = 10^\circ\text{C}$, nhiệt độ tại độ cao x là: $T_x = T_0 - 0.01x$

- Nếu khinh khí cầu được thả ở thời điểm 0, và sau đó bay lên trên với vận tốc 5 mét / giây, hãy vẽ bảng hiển thị các phép đo nhiệt độ và độ cao được báo cáo trong khoảng thời gian 10 giây trong 50 giây đầu tiên. Xác định sai số đo.
- Ở 5000m, nhiệt độ khí cầu gửi về là bao nhiêu?

Dynamic Properties

$$T_r = -0.75e^{-\left(\frac{t}{15}\right)} - 0.05t + 10.75$$

Time	Altitude	Temperature reading	Temperature error
0	0	10	0
10	50	9.86	0.36
20	100	9.55	0.55
30	150	9.15	0.65
40	200	8.70	0.70
50	250	8.22	0.72

Dynamic Properties

□ Second order instrument

If all the coefficients $a_3 \dots a_n$ except for a_0, a_1, a_2 are assumed zero, then:

$$a_2 \frac{d^2 q_0}{dt^2} + \frac{a_1 dq_0}{dt} + a_0 q_0 = b_0 q_i$$

Based on the equation roots, different solutions are obtained:

■ ***Real different roots – over-damped system***

$$\lambda_1 \neq \lambda_2$$
$$q_o(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t} + q_{o\infty}$$

■ ***Real equal roots – critically damped system***

$$\lambda_1 = \lambda_2$$
$$q_o(t) = (C_1 + C_2 t) e^{\lambda_1 t} + q_{o\infty}$$

Dynamic Properties

□ Second order instrument

■ *Complex conjugate pole pair – under-damp system*

$$\lambda_{1,2} = \alpha \pm j\omega$$

$$q_o(t) = (A\cos\omega t + B\sin\omega t)e^{\alpha t} + q_{o\infty}$$

